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i have knowledge on kubernetes and aws eks but never worked on prod env but now for client level interview i need this give a just a list works/tasks done by the engineer in realtime for a application based out of india serving lakhs of users

ChatGPT gpt-6-mini

For an application serving **lakhs of users across India**, an AWS EKS / Kubernetes engineer typically handles the following real-time production responsibilities.

I've arranged these based on actual day-to-day responsibilities, not just Kubernetes concepts.

1. EKS Cluster Management

1. Provision EKS clusters across development, staging and production.
2. Manage managed node groups and worker nodes.
3. Configure EC2 instance types based on application workloads.
4. Upgrade Kubernetes and EKS cluster versions.
5. Perform rolling upgrades of worker nodes.
6. Monitor node health, CPU, memory and disk utilization.
7. Troubleshoot nodes in `NotReady` state.
8. Manage cluster add-ons like CoreDNS, VPC CNI and kube-proxy.
9. Configure cluster autoscaling.
10. Manage cluster access through IAM and Kubernetes RBAC.

2. Application Deployment & Management

1. Deploy applications using Kubernetes Deployments and Services.
2. Manage application replicas based on traffic requirements.
3. Perform rolling deployments without application downtime.
4. Configure readiness, liveness and startup probes.
5. Manage application configurations using ConfigMaps.
6. Manage credentials and sensitive information using Secrets.
7. Configure resource requests and limits for containers.
8. Perform rollback when a new deployment causes issues.
9. Manage different application versions across environments.
10. Troubleshoot CrashLoopBackOff, ImagePullBackOff and Pending pods.

3. EKS Networking & Traffic Management

1. Configure VPC, public and private subnets for EKS.
2. Configure AWS Load Balancer Controller.
3. Expose applications using ALB and Kubernetes Ingress.
4. Configure internal and external load balancing.
5. Manage traffic routing between microservices.
6. Configure security groups and network policies.
7. Troubleshoot pod-to-pod and service-to-service connectivity.
8. Configure Route 53 DNS and application endpoints.
9. Manage NAT Gateway and internet access for private nodes.
10. Monitor load balancer health checks and target groups.

4. Auto Scaling & Performance Management

1. Configure Horizontal Pod Autoscaler (HPA).
2. Configure Cluster Autoscaler or Karpenter for node scaling.
3. Scale application replicas based on CPU and memory utilization.
4. Monitor resource consumption during peak traffic.
5. Perform capacity planning for high-traffic events.
6. Identify resource bottlenecks in application pods.
7. Configure Pod Disruption Budgets.
8. Distribute workloads across availability zones.
9. Configure pod anti-affinity and topology spread constraints.
10. Optimize node utilization and reduce infrastructure costs.

5. Production Monitoring & Troubleshooting

1. Monitor cluster and application health using CloudWatch.
2. Monitor CPU, memory, disk and network utilization.
3. Analyze application and container logs.
4. Troubleshoot pod failures and unexpected restarts.
5. Investigate high response times and application latency.
6. Monitor Kubernetes events and node conditions.
7. Configure Prometheus and Grafana dashboards.
8. Set up alerts for pod failures and resource thresholds.
9. Investigate service downtime and connectivity issues.
10. Participate in production incident management and RCA.

6. Security & Access Management

1. Configure IAM roles for EKS cluster and worker nodes.
2. Implement IRSA or EKS Pod Identity for application access to AWS services.

3. Configure Kubernetes RBAC.
4. Manage Kubernetes Secrets securely.
5. Restrict public access to the EKS API endpoint.
6. Configure private cluster access.
7. Scan container images for vulnerabilities.
8. Implement image security policies.
9. Manage security groups and network restrictions.
10. Apply Kubernetes security best practices.

7. Storage Management

1. Configure persistent volumes and persistent volume claims.
2. Integrate EBS CSI driver for persistent block storage.
3. Configure EFS CSI driver for shared file storage.
4. Troubleshoot volume mounting failures.
5. Manage storage classes.
6. Monitor persistent storage utilization.
7. Handle application data persistence requirements.
8. Configure backup and recovery procedures.

8. CI/CD & Release Management

1. Integrate Jenkins pipelines with EKS.
2. Build Docker images and push them to Amazon ECR.
3. Deploy updated images into Kubernetes clusters.
4. Automate deployments using Helm charts.
5. Maintain Kubernetes YAML manifests.
6. Implement environment-specific deployment configurations.
7. Perform rolling updates and controlled releases.
8. Implement blue-green or canary deployments.
9. Integrate deployment pipelines with Git.
10. Troubleshoot failed deployments and pipeline issues.

9. High Availability & Disaster Recovery

1. Deploy worker nodes across multiple availability zones.
2. Maintain application replicas across different nodes.
3. Configure Pod Anti-Affinity for workload distribution.
4. Configure PDB to maintain application availability during maintenance.
5. Implement automated backup strategies.
6. Plan EKS cluster recovery procedures.
7. Test application recovery after node failures.
8. Maintain infrastructure through Terraform or CloudFormation.

9. Perform disaster recovery drills.
10. Validate application availability during infrastructure maintenance.

10. Real-time scenarios an EKS engineer handles

These are particularly important for your client interview.

Production situation	Engineer's responsibility
Traffic suddenly increases	Scale pods and worker nodes
Application becomes slow	Check CPU, memory, logs and latency
Pod keeps restarting	Investigate logs, events and probes
Worker node crashes	Reschedule workloads onto healthy nodes
Deployment causes downtime	Roll back to previous version
Application is inaccessible	Check Ingress, ALB, Services and networking
Database connectivity fails	Check security groups, endpoints and credentials
Disk usage increases	Investigate node and container storage
New vulnerability discovered	Scan images and deploy patched versions
Entire AZ experiences issues	Maintain service through healthy AZs

11. What a typical engineer does daily

Most relevant for client interviews

Frequency	Typical activities
Daily morning	Check cluster health, nodes, pods and alerts
Daily	Monitor application performance and resource utilization
Daily	Check deployment status and failed pods
Daily	Investigate production tickets and incidents
Daily	Review logs and resolve application issues
As required	Deploy new application versions
As required	Scale workloads during peak traffic
Weekly	Review resource utilization and capacity
Weekly	Review security vulnerabilities and patches
Monthly	Plan upgrades, maintenance and cost optimization

My recommendation: For an engineer with moderate Kubernetes and AWS EKS knowledge, prioritize these six areas for your client interview:

1. EKS cluster and worker node management.
2. Application deployment and troubleshooting.
3. Auto Scaling (HPA + Cluster Autoscaler).
4. Networking (VPC + ALB + Ingress).
5. Monitoring and production incident management.
6. CI/CD and rolling deployments.

These six areas cover a major portion of practical EKS operations.

One important interview tip: Since you haven't actually handled production EKS, don't claim direct production ownership. You can confidently explain the production workflow, architecture, responsibilities and troubleshooting approach as your technical understanding. Be clear about what you've implemented hands-on versus what you understand conceptually.